Math 50 Final: Practice Problems

Instructor: Ethan Levien

Exercise 1: Consider the following code.

```
> n = 100000
> x1 = np.random.normal(0, 2, n)
> x2 = 0.5*x1 + np.random.normal(0, 1, n)
> y = 2*x1 + 3*x2 + np.random.normal(0, 1, n)
```

(a) What is the slope β_1' in the regression with response variable Y and the single predictor variable X_1?

(b) What is the slope β_2' in the regression with response variable Y and the single predictor variable X_2?

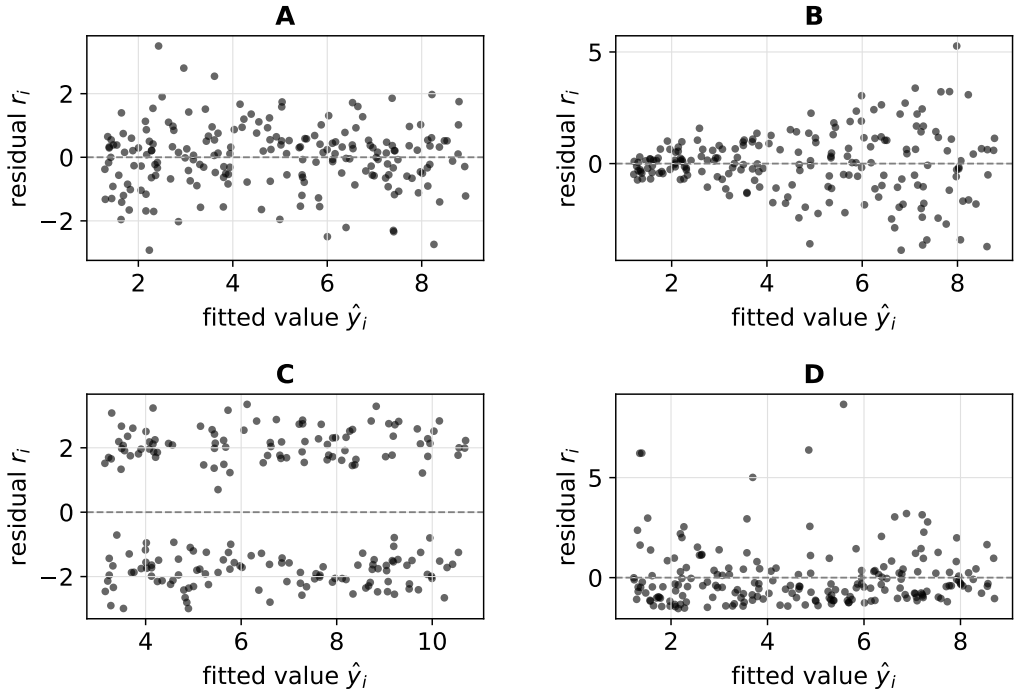
Exercise 2: Consider the following code.

```
> n = 100000
> x = np.random.choice([0, 1, 2], size=n, p=[0.5, 0.3, 0.2])
> y = np.random.normal(1 + 2*x, 1)
> print(np.mean(y))
> print(np.var(y))
> print(np.mean(y[x == 2]))
```

(a) Write the model being simulated in $\sim$ notation, for X and $Y \mid X$.

(b) What (approximately) do the three `print` statements produce?

Exercise 3: A single-predictor model $Y = \beta_0 + \beta_1 X + \epsilon$, with $\epsilon \sim \text{Normal}(0, \sigma_\epsilon^2)$ independent of X, is fit by least squares to four different datasets. The residuals r_i are plotted against the fitted values $\hat{y}_i$:



For each plot, say whether the model's assumptions look reasonable. If not, say which assumption fails and what the data might look like.

Exercise 4: Consider the single-predictor model $Y = \beta_0 + \beta_1 X + \epsilon$ with $\beta_0 = 0$ and $E[X] = 0$, fit to N points with $\hat{\sigma}_X^2 = \frac{1}{N}\sum_{i=1}^{N} X_i^2$ (treat the X_i as fixed). Recall that the least-squares estimate has $\text{var}(\hat{\beta}_1) = \sigma_\epsilon^2/(N\hat{\sigma}_X^2)$. A regularized estimator of β_1 is

$$\hat{\beta}_1^\lambda = \frac{\hat{\beta}_1}{1 + \lambda/\hat{\sigma}_X^2}.$$

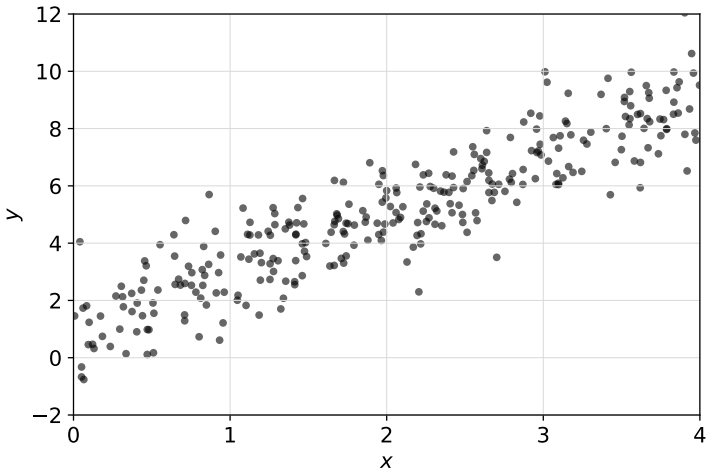
Suppose the true slope is $\beta_1 = 3$.

(a) Compute the bias and the variance of $\hat{\beta}_1^\lambda$ in terms of λ, $\hat{\sigma}_X^2$, σ_ϵ^2 and N.

(b) What is the MSE of $\hat{\beta}_1^\lambda$ when $\lambda = 1$, $\hat{\sigma}_X^2 = 1$, $\sigma_\epsilon^2 = 4$ and $N = 100$?

(c) The true regression function is $f(x) = E[Y \mid X = x] = \beta_1 x$, and the prediction from a training set D is $\hat{y}(x, D) = \hat{\beta}_1^\lambda x$. With the same values as in (b), compute $\text{MSE}_{\hat{y}(2,D)} = E_D\left[(\hat{y}(2, D) - f(2))^2\right]$.

Exercise 5: The plot shows 300 points from a model $Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$.



(a) Estimate β_0 and β_1 from the plot.

(b) Estimate σ_ϵ.

(c) Estimate $\text{var}(X)$.

Exercise 6: Consider the model

$$Y = \beta_0 + \sum_{j=1}^{K} \beta_j \phi_j(X) + \epsilon,$$

with $\beta_0 = 0$, fit by least squares to N points (X_i, Y_i). Treat the X_i as fixed, and suppose the noise terms ϵ_i are independent with variance σ_ϵ^2. The features are uncorrelated and mean zero in the data:

$$\frac{1}{N}\sum_{i=1}^{N} \phi_j(X_i)\phi_k(X_i) = 0 \quad (j \neq k), \qquad \frac{1}{N}\sum_{i=1}^{N} \phi_j(X_i) = 0.$$

(a) Write the least-squares estimate $\hat{\beta}_j$ in terms of the data.

(b) Compute $\text{var}_D(\hat{\beta}_j)$, the variance of the fitted coefficient over the distribution of the data used to fit it. Hint: Since X_i are fixed, the only random variable in there is Y.

Exercise 7: In a class, $X = 1$ if a student did the practice problems (0 otherwise) and $Y = 1$ if the student passed the exam (0 otherwise). The joint distribution is

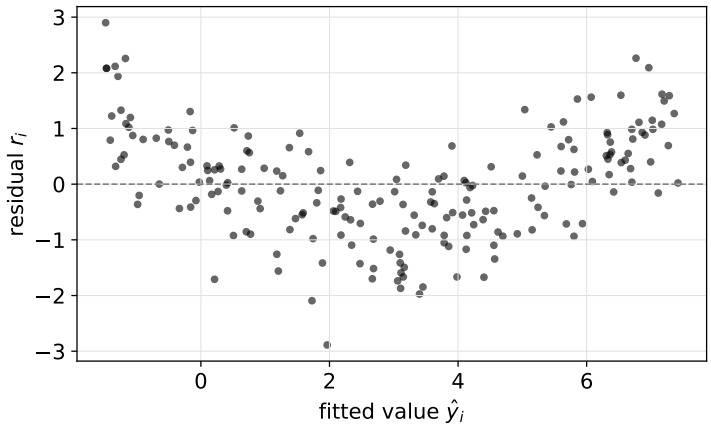
	$Y = 0$	$Y = 1$
$X = 0$	0.4	0.1
$X = 1$	0.2	0.3

(a) Compute $E\left[X^{50}Y^{7}\right]$.

(b) Compute $E\left[(X+Y)^2\right]$.

(c) Compute $\text{var}\left(X^{10}Y^{10}\right)$ and $\text{cov}(X, Y)$.

Exercise 8: A single-predictor model $Y = \beta_0 + \beta_1 X + \epsilon$ is fit to data by least squares. The plot shows the residuals $r_i = Y_i - \hat{y}_i$ against the fitted values $\hat{y}_i$.



Explain what the true relationship between X and Y might be like. Sketch it, together with the fitted line, and say how you would change the model.

Exercise 9: We can view a political system as a way of *estimating*, from citizens' votes, which policies best satisfy some set of criteria. Consider three cases:

- **Direct democracy:** each issue is decided by a popular vote.
- **Representative democracy:** elected officials decide many issues on voters' behalf.
- **Dictatorship:** a small number of individuals make all the decisions.

Discuss the pros and cons of these systems in terms of the bias-variance tradeoff.

Exercise 10: A biologist studies how much plants grow, Y (in cm), as a function of the daily water X_1 (in ml) and fertilizer X_2 (in g) they receive. In her greenhouse, plants that were given more water were usually also given more fertilizer: the correlation between X_1 and X_2 is 0.95. She fits

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

and finds $R^2 = 0.8$, but the 95% confidence intervals for β_1 and β_2 are both very wide.

(a) Sketch what the predictor values $(X_{1,i}, X_{2,i})$ in her data might look like, with X_1 on the horizontal axis and X_2 on the vertical axis. On a second set of axes, sketch predictor values from an experiment that would let her estimate β_1 and β_2 well.

(b) Use your first sketch to explain why β_1 and β_2 are each poorly determined, while the effect of increasing water and fertilizer together, $\beta_1 + \beta_2$, is well determined (take X_1 and X_2 on similar scales).

(c) To decide whether fertilizer matters, she fits a model with X_2 as the only predictor and finds a large slope. Does this show that fertilizer helps the plants grow? Explain.